American Computer Science League

2024-2025 ● Contest 3: Manhattan Distance ● Senior Division

PROBLEM: In a *Sinusoidal Triangle* pattern, numbers are written from left-to-right, then right-to-left, then left-to-right, etc. with decreasing/increasing row widths. The width of rows decreases until there is just one number in the row; the width then increases until reaching the initial width; then decreases, and so on. The diagram below shows the first few rows of a 5-wide Sinusoidal Triangle pattern:

```
 1  2  3  4  5
 9  8  7  6
10 11 12
14 13
15
17 16
18 19 20
24 23 22 21
25 26 27 28 29
33 32 31 30
34 35 36
....
```

The *Manhattan Distance* between two numbers, *M* and *N*, in a Sinusoidal Triangle is how far the numbers are from each other in both the horizontal and vertical directions. You'll be given the width of a Sinusoidal Triangle pattern, and two numbers, *M* and *N,* in the pattern. Your program must find the Manhattan Distance between numbers *M* and *N*.

EXAMPLE:

Input	`5 17 2`
Output	`6`
Explanation	Build the pattern as shown above. The Manhattan Distance between 17 and 2 is 6: go horizontally 1 column and vertically 5 rows to get from one number to the other.

INPUT: Three positive integers: *width*, *M*, and *N*, each no more than 1,000.

OUTPUT: Find the Manhattan Distance between *M* and *N* in the Sinusoidal Triangle with the specified initial *width.*

SAMPLE INPUT *(for readability, the 3 integer inputs are shown on a single line):*

```
1. 5 17 2
2. 3 24 25
3. 7 40 30
4. 6 5 100
5. 8 200 180
6. 4 100 120
```

SAMPLE OUTPUT:

```
1. 6
2. 1
3. 5
4. 29
5. 4
6. 8
```

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TEST DATA

TEST INPUT *(for readability, the 3 integer inputs are shown on a single line):*

```
1. 5 44 33
2. 4 20 70
3. 6 100 200
4. 3 20 28
5. 10 1000 200
6. 25 1000 950
```

TEST OUTPUT:

```
1. 7
2. 22
3. 31
4. 4
5. 152
6. 10
```

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PROBLEM STATEMENT:
In a *Sinusoidal Triangle* pattern, numbers are written from left-to-right, then right-to-left, then left-to-right, etc. with decreasing/increasing row widths. The width of rows decreases until there is just one number in the row; the width then increases until reaching the initial width; then decreases, and so on. The diagram below shows the first few rows of a 5-wide Sinusoidal Triangle pattern:

```
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. . . .
```

The *Manhattan Distance* between two numbers, *M* and *N*, in a Sinusoidal Triangle is how far the numbers are from each other in both the horizontal and vertical directions. You'll be given the width of a Sinusoidal Triangle pattern, and two numbers, *M* and *N,* in the pattern. Your program must find the Manhattan Distance between numbers *M* and *N*.

EXAMPLE:
Input:
```
5 17 2
```

Output:
```
6
```

Explanation:
Build the pattern as shown above. The Manhattan Distance between 17 and 2 is 6: go horizontally 1 column and vertically 5 rows to get from one number to the other.

TASK:
Complete the function **findManDist**

- The function has 3 parameters, all positive integers: *width*, the maximum width of the longest rows in the pattern, and the two numbers, *M* and *N*, representing the 2 numbers to use in finding the Manhattan Distance.
- The function returns an integer for the Manhattan Distance between *M* and *N* in the resulting pattern.

You may create additional functions that are called from **findManDist** if needed in solving the problem.

CONSTRAINTS:
Each input value is no more than 1,000.

DATA PROVIDED:
There are 6 sets of Sample Data for debugging and 6 sets of Test Data for scoring. You may create additional data sets for debugging your program.

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问题：在*正弦波三角形*模式中，数字按照从左至右、从右至左、再从左至右的顺序依次书写，行宽交替递减或递增。行宽逐渐缩小，直至某一行仅剩一个数字；随后行宽逐渐扩大，直至恢复到初始宽度；之后再次缩小，如此循环往复。**下图展示了一个**宽度**为** 5 **的正弦**波**三角形模式的前几行：**

```
 1  2  3  4  5
 9  8  7  6
10 11 12
14 13
15
17 16
18 19 20
24 23 22 21
25 26 27 28 29
33 32 31 30
34 35 36
....
```

在正弦波**三角形中，两个数字** M **和** N **之间的*曼哈顿距离*是指这两个数字在水平和垂直方向上的距离之和。题目会给出一个正弦**波**三角形模式的**宽度**以及该模式中的两个数字**M **和** N**。你的程序需要计算数字** M **和** N **之间的曼哈顿距离。**

样例：

输入	5 17 2
输出	6
*解释**说明***	按上**图所示**构建模式。17 和 2 之间的曼哈顿距离为 6：从 17 到 2 **需要**水平移动 1 列，垂直移动 5 行。

输入：三个正整数：*宽度*、M **和** N**，每个数均不超过** 1000**。**

输出：在指定*初始宽度*的正弦波**三角形中，计算** M **和** N **之间的曼哈顿距离。**

样例输入（*为便于阅读，输入的 3 个整数均显示在同一行*）：

```
1. 5 17 2
2. 3 24 25
3. 7 40 30
4. 6 5 100
5. 8 200 180
6. 4 100 120
```

样例输出：

```
1. 6
2. 1
3. 5
4. 29
5. 4
6. 8
```

测试数据

测试输入（*为便于阅读，输入的 3 个整数均显示在同一行*）：

```
1. 5 44 33
2. 4 20 70
3. 6 100 200
4. 3 20 28
5. 10 1000 200
6. 25 1000 950
```

测试输出：

```
1. 7
2. 22
3. 31
4. 4
5. 152
6. 10
```

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问题陈述：

在正弦波三角形模式中，数字按照从左至右、从右至左、再从左至右的顺序依次书写，行宽交替递减或递增。行宽逐渐缩小，直至某一行仅剩一个数字；随后行宽逐渐扩大，直至恢复至初始宽度；之后再次缩小，如此循环往复。**下图展示了一个**宽度**为** 5 **的正弦**波**三角形模式的前几行：**

```
 1  2  3  4  5
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. . . .
```

在正弦波**三角形中，两个数字** M **和** N **之间的*曼哈顿距离*是指这两个数字在水平和垂直方向上的距离之和。题目会给出一个正弦**波**三角形模式的**宽度**以及该模式中的两个数字** M **和** N**。你的程序需要计算数字** M **和** N **之间的曼哈顿距离。**

样例：

输入：

```
5 17 2
```

输出：

```
6
```

*解释**说明：***

按上**图所示**构建模式。17 和 2 之间的曼哈顿距离为 6：从 17 到 2 **需要**水平移动 1 列，垂直移动 5 行。

任务：

完成函数 findManDist

- **该函数有** 3 **个参数，均为正整数：***宽度***表示模式中最长行的**最大宽度**，** M **和** N **表示用于计算曼哈顿距离的两个数字。**

- **函数返回一个整数，表示在**生成**的模式中，** M **和** N **之间的曼哈顿距离。**

在解决问题的过程中，如有需要，你可以构建额外的函数供 findManDist **调用来解答此问题。**

约束条件：

每个输入值不超过 1000**。**

提供的数据：

有 6 **组用于调试的样例数据和** 6 **组用于评分的测试数据。你应该创建自己的样例数据来充分测试你的程序。**